

Assignment 2 Report Template: Domain Adaptation with Small Language Models

Group Number: *Insert Group Number*
CSCI433/933 Machine Learning Algorithms and Applications
Student Names and Student Numbers: *Insert Details*

Main Report Page Limit: The main report must not exceed **10 pages**, in the style of IEEE conference submissions. The 10-page (absolute) limit applies from the abstract through the conclusion. References, appendices, full evaluation tables, extended prompts, and the GenAI usage log are excluded from the 10-page limit. Marks are awarded for clarity, technical depth, evidence, and quality of reasoning, not for length.

Abstract—Briefly summarise the problem, the system developed, the model/RAG approach used, the dataset, the evaluation method, and the main findings. This should be approximately 150–200 words.

I. INTRODUCTION AND PROBLEM FORMULATION

Relevant marks: Problem formulation and task understanding (10 marks).

Explain the purpose of the project. Describe the task as a domain adaptation problem involving a pretrained Small Language Model (SLM), Shakespearean text, retrieval-augmented generation, and beginner-friendly explanation.

Your introduction should address:

- what problem your system is trying to solve;
- why Shakespearean text presents a domain adaptation challenge;
- who the intended user is;
- what your system is expected to do;
- what constraints shaped your design.

Avoid writing a long generic literature review. Focus on the specific problem, system objective, and assignment requirements.

II. DATASET AND PREPROCESSING

Relevant marks: Dataset preparation and documentation (15 marks).

Describe the dataset used in the project. At minimum, explain how the three required plays were represented and prepared:

- *Hamlet*;
- *Macbeth*;
- *Romeo and Juliet*.

You should describe:

- the structure of the dataset;
- whether you used utterance-level, scene-level, or other chunking;

- metadata retained, such as play, act, scene, speaker, and source identifier;
- any cleaning, filtering, or normalisation performed;
- any supplementary material used, such as summaries or glossaries;
- how primary text was distinguished from supporting explanatory material.

A. Dataset Schema

Provide a concise description of your processed data format. For example:

```
{
  "play": "Macbeth",
  "act": 1,
  "scene": 3,
  "speaker": "MACBETH",
  "text": "So foul and fair a day I have not seen.",
  "source_id": "macbeth_1_3_macbeth_001"
}
```

B. Chunking Strategy

Explain your chunking strategy and justify why it is suitable for retrieval. Discuss any trade-offs between short chunks, which may be precise but lack context, and longer chunks, which may preserve context but reduce retrieval specificity.

III. METHODOLOGY

Relevant marks: Model adaptation and RAG methodology (20 marks).

Describe your technical approach. This section should explain both your baseline system and your improved RAG-based system.

A. Baseline System

Describe the baseline system used for comparison. This may be a prompt-only model, a simple retrieval-only system, or another minimal approach.

You should explain:

- what the baseline does;
- what model or tool it uses;
- what information it has access to;
- why it is a fair baseline for comparison.

B. RAG-Based System

Describe your retrieval-augmented generation pipeline. At minimum, explain:

- embedding model used;
- retrieval/indexing method used;
- number of retrieved chunks used for generation;
- prompt structure;
- language model used for generation;
- how retrieved context is incorporated into the generated answer.

C. Model Choice and Justification

Justify your choice of model. Discuss how your approach reflects the principles of small, deployable, or resource-constrained language-model systems.

If you used a hosted model, explain why this was appropriate and how the rest of your system design still reflects the assignment’s SLM constraints.

IV. SYSTEM DESIGN AND IMPLEMENTATION

Relevant marks: System implementation and usability (15 marks).

Describe the implemented system clearly enough that a reader can understand how the components interact.

Include a system diagram if useful.

A. System Pipeline

Describe the end-to-end flow:

- 1) user query;
- 2) query embedding;
- 3) retrieval of relevant passages;
- 4) prompt construction;
- 5) response generation;
- 6) evidence display;
- 7) logging or evaluation output.

B. Implementation Details

Summarise the main implementation details:

- programming language and libraries;
- repository structure;
- how to run the system;
- dependencies;
- any cached embeddings or indices;
- limitations of the implementation.

Do not paste large blocks of code in the main report. Place important snippets only where they support explanation. Long code fragments should be placed in the repository, not in the report.

V. EVALUATION DESIGN

Relevant marks: Evaluation, comparison, and error analysis (20 marks).

Describe your evaluation protocol.

A. Evaluation Questions

State how many questions were used and distinguish between:

- instructor-provided questions;
- group-designed questions.

Briefly explain what your group-designed questions were intended to test.

B. Scoring Criteria

Define the scoring scale used for each criterion. For example, if you use a 1–5 scale, define what scores 1, 3, and 5 mean.

Your evaluation should include:

- correctness;
- grounding;
- retrieval relevance;
- usefulness for a beginner;
- style quality, where applicable.

C. Evaluation Table Format

A compact summary table may be included in the main report. Full detailed evaluation tables should be placed in the appendix.

TABLE I
EXAMPLE COMPACT EVALUATION SUMMARY.

System	Correctness	Grounding	Usefulness
Baseline	<i>score</i>	<i>score</i>	<i>score</i>
RAG System	<i>score</i>	<i>score</i>	<i>score</i>

VI. RESULTS AND ANALYSIS

Relevant marks: Evaluation, comparison, and error analysis (20 marks).

Present and interpret the results. Do not merely list outputs. Explain what the results show about the strengths and weaknesses of your system.

A. Baseline versus RAG Comparison

Compare the baseline system with the improved RAG-based system.

Discuss:

- whether RAG improved factual correctness;
- whether retrieved evidence improved answer grounding;
- cases where RAG helped;
- cases where RAG did not help;
- any unexpected behaviour.

B. Qualitative Examples

Include a small number of representative examples. For each example, show:

- the user question;
- retrieved evidence;
- generated response;

- your interpretation of the result.

Keep examples concise in the main report. Longer examples should be placed in the appendix.

C. Failure Analysis

Discuss at least three failure cases. Possible failure types include:

- irrelevant retrieval;
- incomplete retrieval;
- hallucinated answer;
- over-stylised or unclear response;
- correct answer but weak evidence;
- beginner-unfriendly explanation;
- confusion between plays or characters.

For each failure case, explain what went wrong and how it might be improved.

VII. RESPONSIBLE USE OF GENERATIVE AI

Relevant marks: Responsible and transparent GenAI usage (10 marks).

Summarise how generative AI tools were used in this project.

You should describe:

- what tools were used;
- what they were used for;
- how outputs were verified;
- what was accepted, modified, rejected, or improved;
- how the group ensured that submitted work reflects its own understanding.

The full GenAI usage log may be placed in an appendix or submitted as a separate file.

VIII. CONCLUSION

Summarise what was achieved, what was learned, and what limitations remain.

Your conclusion should address:

- whether the system met the assignment objectives;
- what the evaluation revealed;
- what design choices were most important;
- what you would improve with more time.

REFERENCES

[1] Replace this placeholder with the references used in your report. Students may alternatively use BibTeX by replacing this environment with: `\bibliographystyle{IEEEtran}` and `\bibliography{references}`.

APPENDIX A

FULL EVALUATION TABLE

Include the full evaluation results here.

APPENDIX B

REPRESENTATIVE PROMPTS

Include important prompts used for retrieval, answer generation, style control, and evaluation.

[Insert system prompt or RAG prompt here.]

APPENDIX C

ADDITIONAL OUTPUT EXAMPLES

Include additional examples that support your analysis but do not fit within the 10-page main report.

APPENDIX D

GENAI USAGE LOG

The GenAI usage log should include enough information to make the group's use of AI transparent and assessable.

TABLE II
FULL EVALUATION TABLE TEMPLATE.

Question	Expected Focus	Correctness	Grounding	Usefulness	Comments
<i>Insert question</i>	<i>Insert focus</i>	<i>Score</i>	<i>Score</i>	<i>Score</i>	<i>Comment</i>

TABLE III
GENAI USAGE LOG TEMPLATE.

Tool	Prompt or Task	Useful Output	Verification / Modification
<i>Tool name</i>	<i>Prompt summary</i>	<i>Response summary</i>	<i>How it was checked or modified</i>